

# RCP/RCLM Notation Table

| Symbol                                | Meaning                                            | Role                                   |
|---------------------------------------|----------------------------------------------------|----------------------------------------|
| RCP                                   | Recursive Coherence Preservation                   | architecture-general theorem framework |
| RCLM                                  | Recursive Coherence Language Model                 | architecture instantiation             |
| $K_t^{\{SV\text{-seedlib}, N\}}$      | finite SV seed-library domain                      | theorem-entry domain                   |
| $K_t^{\{SV, N\}}$                     | finite successor-verification domain               | trajectory domain                      |
| SVPkg <sub>t</sub>                    | successor-verification packet                      | verification certificate packet        |
| SVBuilder <sub>t</sub>                | successor-verification packet builder              | constructs SV packets                  |
| PCS <sub>t</sub>                      | proof-carrying successor packet                    | reference/checker object               |
| Check <sub>RCP</sub>                  | trusted checker                                    | checks canonical finite packets        |
| BuildRefSV <sub>{0:N}</sub>           | reference builder/compiler                         | constructs finite reference instance   |
| RefSV <sub>{0:N}^{\{RCP/RCLM\}}</sub> | finite reference instance                          | math/architecture witness              |
| MechCert                              | external mechanization certificate                 | Lean 4 scope marker                    |
| LECert                                | learned-entry certificate                          | M3-Min entry boundary                  |
| RealCont                              | reality containment / misspecification certificate | real-world reliability qualifier       |
| SVTract                               | restricted tractability certificate                | strong tractability qualifier          |